

Question 7 - Stage 1

09 : 19 : 29 and 12 : 11 : 10 is each an example of what we will call an Arithmetic Time. These are times in the 24 hour clock, $H : M : S$, when the integers H, M, S , in that order, form an arithmetic sequence with a non-zero common difference.

On an analogue clock, let

$0^\circ \leq \alpha < 180^\circ$ be the angle between the hour and minute hands,

$0^\circ \leq \beta < 180^\circ$ be the angle between the minute and second hands, and

$0^\circ \leq \gamma < 180^\circ$ be the angle between the second and hour hands.

What is the minimum value of $\alpha + \beta + \gamma$ (in degrees to 1 d.p.) for an Arithmetic Time?

Working

Finding all Arithmetic Times

In [1]:

```
from itertools import product

arithmeticTimes = []
totals = []

for time in product( list(range(24)) , list(range(60)) , list(range(60)) ):
    h,m,s = time
    if m-s == h-m and m-s != 0:
        arithmeticTimes.append(time)

print(arithmeticTimes)
```

[(0, 1, 2), (0, 2, 4), (0, 3, 6), (0, 4, 8), (0, 5, 10), (0, 6, 12), (0, 7, 14), (0, 8, 16), (0, 9, 18), (0, 10, 20), (0, 11, 22), (0, 12, 24), (0, 13, 26), (0, 14, 28), (0, 15, 30), (0, 16, 32), (0, 17, 34), (0, 18, 36), (0, 19, 38), (0, 20, 40), (0, 21, 42), (0, 22, 44), (0, 23, 46), (0, 24, 48), (0, 25, 50), (0, 26, 52), (0, 27, 54), (0, 28, 56), (0, 29, 58), (1, 2, 3), (1, 3, 5), (1, 4, 7), (1, 5, 9), (1, 6, 11), (1, 7, 13), (1, 8, 15), (1, 9, 17), (1, 10, 19), (1, 11, 21), (1, 12, 23), (1, 13, 25), (1, 14, 27), (1, 15, 29), (1, 16, 31), (1, 17, 33), (1, 18, 35), (1, 19, 37), (1, 20, 39), (1, 21, 41), (1, 22, 43), (1, 23, 45), (1, 24, 47), (1, 25, 49), (1, 26, 51), (1, 27, 53), (1, 28, 55), (1, 29, 57), (1, 30, 59), (2, 1, 0), (2, 3, 4), (2, 4, 6), (2, 5, 8), (2, 6, 10), (2, 7, 12), (2, 8, 14), (2, 9, 16), (2, 10, 18), (2, 11, 20), (2, 12, 22), (2, 13, 24), (2, 14, 26), (2, 15, 28), (2, 16, 30), (2, 17, 32), (2, 18, 34), (2, 19, 36), (2, 20, 38), (2, 21, 40), (2, 22, 42), (2, 23, 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Finding Smallest Angle

In [2]:

```
for time in arithmeticTimes:
    h,m,s = time

    sProp = s/60
    mProp = m/60 + sProp/60
    hProp = (h/12 if h<12 else (h-12)/12) + mProp/12

    hAngle,mAngle,sAngle = (i * 360 for i in [hProp,mProp,sProp]) # clockwise Angle from Top

    hTm = abs(hAngle-mAngle)
    mTs = abs(mAngle-sAngle)
    sTh = abs(sAngle-hAngle)

    alpha,beta,gamma = ((angle if angle < 180 else 360-angle) for angle in [hTm,mTs,sTh])

    totals.append([alpha + beta + gamma,time])

totals.sort(key=lambda x: x[0])
```

```
minimum = totals[0]
print(f"Smallest time: {minimum[1][0]}:{minimum[1][1]}:{minimum[1][2]}")
print(f"Smallest Total Angle: {round(totals[0][0],1)}")
```

Smallest time: 15:16:17
Smallest Total Angle: 8.6